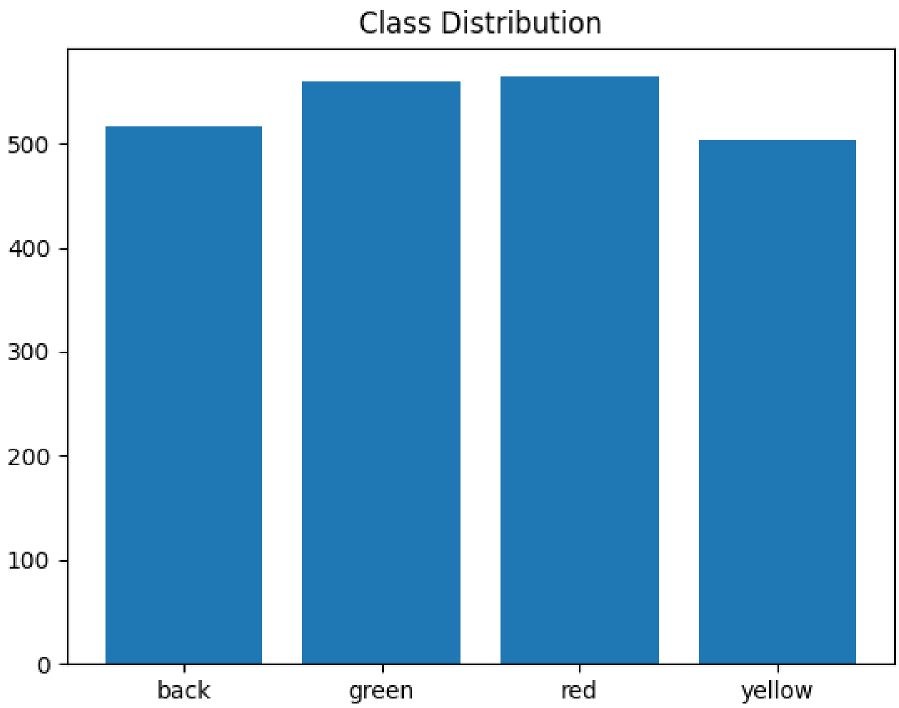
Traffic Light AI - EDA Report

Dataset Classes: back, green, red, yellow

Class Distribution



Sample Images

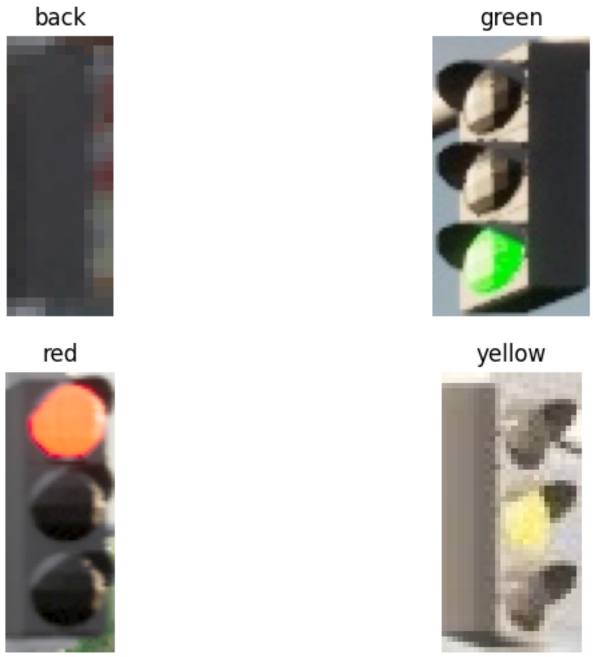

back
green
red
yellow

Image Sizes

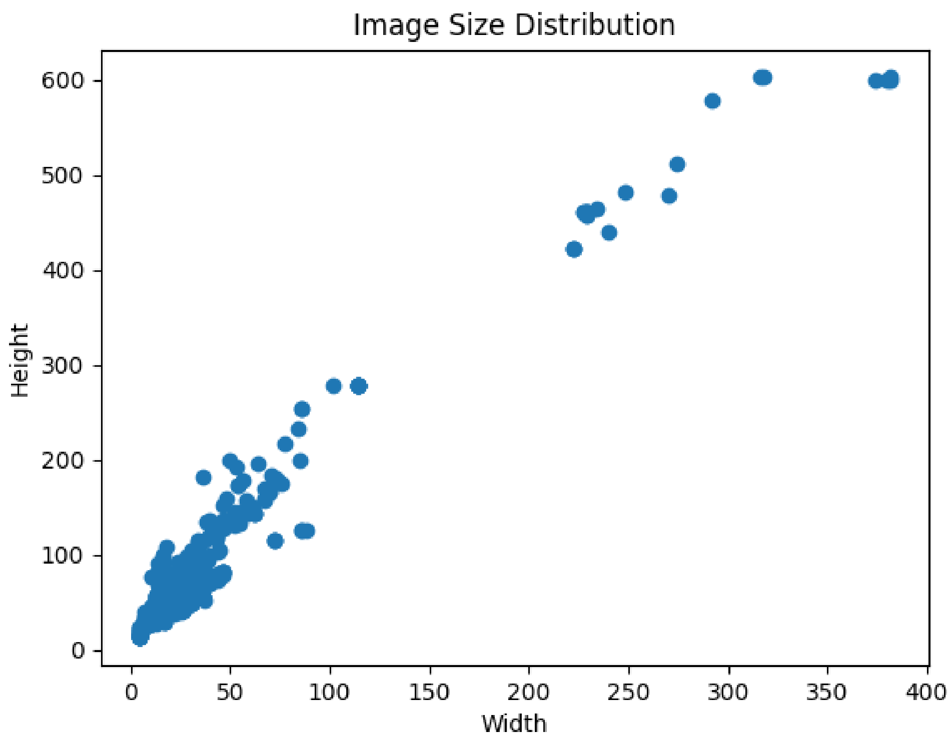
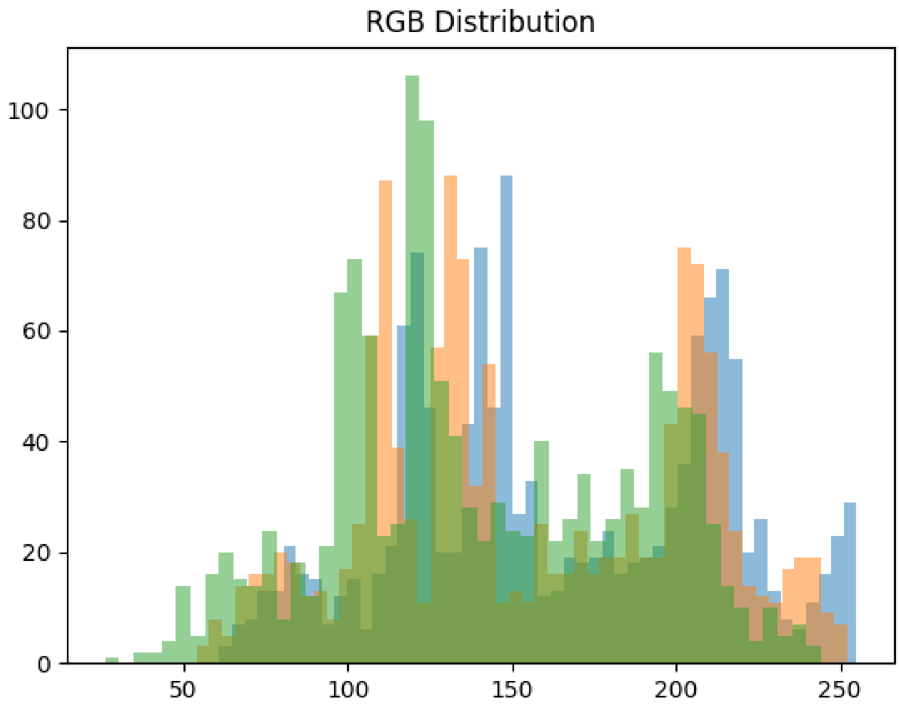


Image Size Distribution
Height
Width
0
100
200
300
400
500
600
0
50
100
150
200
250
300
350
400

Rgb Distribution



EDA Code:

```
import os
import matplotlib.pyplot as plt
from PIL import Image
import numpy as np
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Image as RLImage
from reportlab.lib.styles import getSampleStyleSheet

# Dataset path
DATASET_PATH = "traffic_light_data/train"
OUTPUT_DIR = "outputs"
os.makedirs(OUTPUT_DIR, exist_ok=True)

classes = os.listdir(DATASET_PATH)

print("Classes Found:", classes)

# ------------------------------
# 1■■ CLASS DISTRIBUTION GRAPH
# ------------------------------
class_counts = []
for cls in classes:
count = len(os.listdir(os.path.join(DATASET_PATH, cls)))
class_counts.append(count)

plt.figure()
plt.bar(classes, class_counts)
```

```
plt.title("Class Distribution")
plt.savefig(f"{OUTPUT_DIR}/class_distribution.png")
plt.close()

# -------------------------------
# 2■■ SAMPLE IMAGES
# -------------------------------
plt.figure(figsize=(8,6))
i = 1
for cls in classes:
img_path = os.path.join(DATASET_PATH, cls, os.listdir(os.path.join(DATASET_PATH, cls))[0])
img = Image.open(img_path)
plt.subplot(2,2,i)
plt.imshow(img)
plt.title(cls)
plt.axis("off")
i += 1

plt.savefig(f"{OUTPUT_DIR}/sample_images.png")
plt.close()

# -------------------------------
# 3■■ IMAGE SIZE DISTRIBUTION
# -------------------------------
widths, heights = [], []

for cls in classes:
folder = os.path.join(DATASET_PATH, cls)
for img_name in os.listdir(folder)[:100]:
img = Image.open(os.path.join(folder, img_name))
w,h = img.size
widths.append(w)
heights.append(h)

plt.figure()
plt.scatter(widths, heights)
plt.title("Image Size Distribution")
plt.xlabel("Width")
plt.ylabel("Height")
plt.savefig(f"{OUTPUT_DIR}/image_sizes.png")
plt.close()

# -------------------------------
# 4■■ RGB DISTRIBUTION
# -------------------------------
img = Image.open(img_path)
img_arr = np.array(img)

plt.figure()
plt.hist(img_arr[:,:,0].ravel(), bins=50, alpha=0.5)
plt.hist(img_arr[:,:,1].ravel(), bins=50, alpha=0.5)
plt.hist(img_arr[:,:,2].ravel(), bins=50, alpha=0.5)
plt.title("RGB Distribution")
plt.savefig(f"{OUTPUT_DIR}/rgb_distribution.png")
plt.close()
```

```
print("Graphs saved in outputs folder!")

# -------------------------------
# 5■■ CREATE PDF REPORT
# -------------------------------
doc = SimpleDocTemplate("EDA_Report.pdf")
styles = getSampleStyleSheet()
story = []

story.append(Paragraph("Traffic Light AI - EDA Report", styles['Title']))
story.append(Spacer(1,20))

story.append(Paragraph("Dataset Classes: " + ", ".join(classes), styles['Normal']))
story.append(Spacer(1,20))

images = [
"class_distribution.png",
"sample_images.png",
"image_sizes.png",
"rgb_distribution.png"
]

for img in images:
story.append(Paragraph(img.replace("_"," ").replace(".png","").title(), styles['Heading2']))
story.append(RLImage(f"{OUTPUT_DIR}/{img}", width=400, height=300))
story.append(Spacer(1,20))

# -------------------------------
# 6■■ ADD CODE IN PDF
# -------------------------------
code_text = open("eda.py", encoding="utf-8").read().replace("\n","
")
story.append(Paragraph("EDA Code:", styles['Heading1']))
story.append(Paragraph(code_text, styles['Normal']))
story.append(Spacer(1,20))

# -------------------------------
# 7■■ SUMMARY (VERY IMPORTANT)
# -------------------------------
summary = """
• Dataset contains 4 classes: Red, Green, Yellow, Background.
• Class distribution checked to ensure balance.
• Sample images verified visually.
• Image size distribution analyzed.
• RGB distribution analyzed to understand color patterns.
• Dataset is suitable for CNN training.
"""

story.append(Paragraph("EDA Summary:", styles['Heading1']))
story.append(Paragraph(summary, styles['Normal']))

doc.build(story)

print("EDA_Report.pdf Generated Successfully!")
```

EDA Summary:

• Dataset contains 4 classes: Red, Green, Yellow, Background. • Class distribution checked to ensure balance. • Sample images verified visually. • Image size distribution analyzed. • RGB distribution analyzed to understand color patterns. • Dataset is suitable for CNN training.